

RM6 is an indoor gas-insulated switchgear up to 24kV for secondary distribution networks.



RM6 meets the definition of a "sealed pressure system" as laid out by the IEC standard.

RM6 is made up of the following elements:

1. A stainless steel tank filled with SF₆ gas (at 0.23 bar relative pressure), sealed for life and containing the busbar and all live switching components such as the switch-disconnector, the earthing switch, the fuse switch combination or the circuit breaker
2. One to four (five optional) cable compartments with interfaces to connect to the network or the transformer
3. User interface with single line diagram, actuators and LV components
4. Manual or motorized operating mechanism compartments
5. Earthing circuit with visible earthing contacts

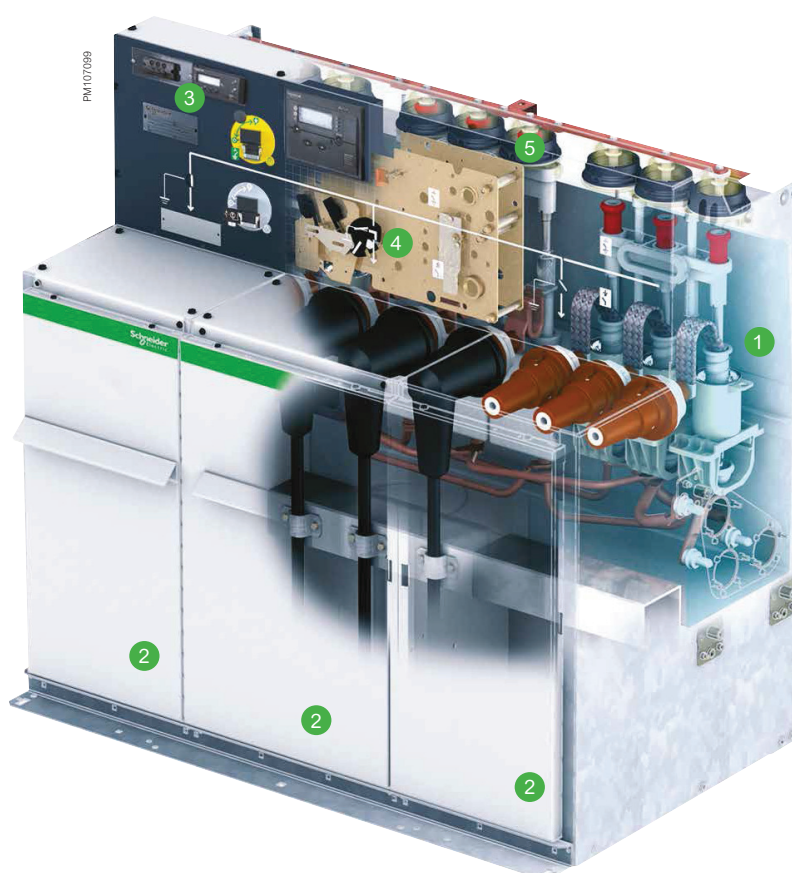
Electrical characteristics

Rated voltage	Ur (kV)	12	17.5	24
Frequency	f (Hz)	50 or 60		
Insulation level				
Industrial frequency	Insulation (1) Ud (kV rms)	28	38	50
50Hz 1 mn	Isolation (2) Ud (kV rms)	32	45	60
Impulse 1.2/50 μs	Insulation (1) Up (kV peak)	75	95	125
	Isolation (2) Up (kV peak)	85	110	145
Tank internal arc withstand	20kA 1s			
Seismic Withstand	Severity class 2, acceptance class 2 as per IEC62271-210 (2013)			
Vibration Withstand	NF EN60068.2.6.2 (2008) (3)			

(1) Phase-to-phase, phase-to-earth

(2) Across the isolating distance

(3) Please contact Schneider Electric for details



Operating conditions and standards



RM6 performances meet the definition of a “sealed pressure system” as laid down in the IEC recommendations.

The RM6 tank is filled with SF6 at 0.23bar relative pressure and sealed for life after filling. Its tightness, which is systematically checked at the factory, gives the switchgear a high life expectancy.

RM6 is designed in accordance with the following IEC standards used for general operation conditions for indoor switchgear:

IEC 62271-1 (common specifications for high voltage switchgear and controlgear)

Ambient temperature: class –25°C indoor

- Lower than or equal to 40°C without derating
- Lower than or equal to 35°C over 24 hours average without derating
- Greater than or equal to –25°: please contact us for details

Altitude:

- Lower than or equal to 1000 m
- Above 1 000 m, and up to 2 000 m with direct field connectors
- Greater than 2 000 m: please contact us for further details

DE-Mt needs voltage derating after 1 000 m.

Please consider altitude and temperature when selecting Q function fuses.

Current derating according to ambient temperature						
	(°C)	40	45	50	55	60
Busbars 630 A	Ir (A)	630	575	515	460	425
Busbars 400 A	Ir (A)	400	400	400	355	
Functions: I, O, B (with bushing type C)	(A)	630	575	515	460	425
Function D (with bushing type B or C)	(A)	200	200	200	200	200
Function Q	(A)	(3)	(4)	(4)	(4)	(4)

(3) Depends on fuse selection

(4) Please contact us

Operating conditions and standards



IEC 62271-200

(AC metal enclosed switchgear and controlgear for rated voltage above 1 kV and up to and including 52 kV)

- Switchgear classification: PM class (metallic partitioning)
- Loss of service continuity: LSC2 class
- Internal arc classification up to A-FLR 20kA 1 sec. (Please refer to section referring to internal arc performance for precise values)

Switch-disconnectors

IEC 62271-103 (high voltage switches for rated voltage above 1 kV and less than 52 kV)

- Class M1/E3
- 100 CO cycles at rated current and 0.7 p.f.
- 1000 mechanical opening operations.

Circuit breakers: 200 A feeder or 630 A line protection

IEC 62271-100 (high voltage alternating current circuit breakers)

- Class M1/E2
 - 2000 mechanical opening operations,
 - O-3 min.-CO-3 min.-CO cycle at rated short circuit current

Other applicable standards

IEC 62271-100 (high voltage alternating current circuit breakers)

- Switch-fuse combinations: IEC 62271-105: alternating current switch-fuse combination.
- Earthing switch: IEC 62271-102: alternating current disconnectors and earthing switches.
- Electrical relays: IEC 60255.

RM6 Protection Index

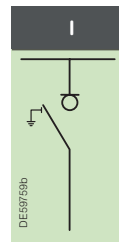
- Tank with HV parts: IP67
- Front face + mechanism: IP3X
- Protection against mechanical impact: IK07

Functional overview

I, Ic functions

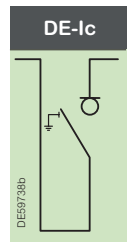
I function

- Network points with switch-disconnector



DE-Ic function

- Bus coupler by switch-disconnector



Rated voltage			Ur	kV	12	17,5	24		
Rated frequency			Fr	Hz	50 or 60	50 or 60	50 or 60		
Insulation level									
Industrial frequency 50Hz/1min	Phase-to-phase, phase-to-earth	Ud	kV rms	28	38	50			
	Across isolating distance	Ud	kV rms	32	45	60			
Lightning impulse withstand	Phase-to-phase, phase-to-earth	Up	kV peak	75	95	125			
	Across isolating distance	Up	kV peak	85	110	145			
Rated current		Ir	A	630	630	400	400	630	630
Rated current busbars		Ir	A	630	630	400	400	630	630
Rated peak current		Ip	kA	62.5	52.5	31.25	40	40	50
Short-time withstand current		It	kA rms	25	21	12.5	16	16	20
		tk	s	1	1 or 3	1	1	1	1 or 3
Breaking capacity	Active load	Iload	A	630	630	400	400	630	630
	Earth fault	Ief1	A	320	320	320	320	320	320
	Cable charging	Icc	A	110	110	110	110	110	110
Making capacity of switch and earthing switches		Ima	kA peak	62.5	52.5	31.25	40	40	50
Bushing (1)			Type	C	C	B or C	B or C	C	C
Mechanical endurance	Switch-disconnector	M1	Number of openings	1000	1000	1000			
	Earthing switch	M0	Number of openings	1000	1000	1000			
Electrical endurance	Switch-disconnector	E3	Number of CO at rated current	100	100	100			
			Number of short-circuit making operations	5	5	5	5	5	2
	Earthing switch	E2	Number of CO at rated current	100	100	100			
			Number of short-circuit making operations	5	5	5	5	5	2

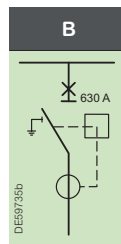
(1) No bushing for IC function

Functional overview

B, D, BC functions

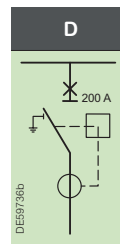
B function

- Network points with 630 A disconnecting circuit breaker (line protection feeder)



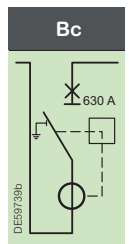
D function

- Transformer feeder 200 A with disconnecting circuit breaker



DE-Bc function

- Bus coupler by 630 A circuit breaker



Rated voltage		Ur	kV	12		17,5		24						
Rated frequency		Fr	Hz	50 or 60		50 or 60		50 or 60						
Insulation level														
Industrial frequency 50Hz/1min	Phase-to-phase, phase-to-earth	Ud	kV rms	28		38		50						
	Across isolating distance	Ud	kV rms	32		45		60						
Lightning impulse withstand	Phase-to-phase, phase-to-earth	Up	kV peak	75		95		125						
	Across isolating distance	Up	kV peak	85		110		145						
Rated current		Ir	A	200	630	200	630	200	630	200	630	200	200	200
Rated current busbars		Ir	A	630		630		630		630		400	400	630
Short-time withstand current		It	kA rms	25		21(1)		16		20		12.5	16	12.5
		tk	s	1		1 or 3		1		1 or 3		1	1	1
No-load transformer breaking capacity		I3	A	-	16	-	16	-	16	-	16	16	16	16
Short-circuit breaking capacity		Isc	kA	25		21		16		20		12.5	16	12.5
Making capacity		Ima	kA peak	62.5		52.5		40		50		31.25	40	31.25
Operating sequence				O – 3min- CO – 3min - O										
Bushing(2)			Type	C		C		C		C		A	BorC	A
Mechanical endurance	Circuit breaker	M1	Number of openings	2000		2000		2000						
	Earthing switch	M0	Number of openings	1000		1000		1000						
Electrical endurance	Circuit breaker	E2	Number of short-circuit breaking operations	3		3		3						
			Number of short-circuit making operations	2		2		2						
	Earthing switch	E2	Number of CO at rated current	100		100		100						
			Number of short-circuit making operations	5		5		5		2		5	5	5

(1) 17.5 kA for DE-Bc

(2) No bushing for DE-Bc function